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Research Article

Computational Characterization of Natural Compounds as Selective Inhibitors of Mutant NRAS in Melanoma

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ABSTRACT

Melanoma is an aggressive and currently incurable skin cancer, primarily affecting individuals with light skin pigmentation. Key oncogenic drivers include mutations in BRAF, NRAS, CDK4, and MITF, with signaling mediated mainly through the MAPK and PI3K–Akt pathways. MEK inhibitors have shown limited clinical success, emphasizing the need for novel therapeutic agents. In this study, structure-based virtual screening and molecular docking were employed to identify natural products that inhibit mutant NRAS. The crystal structure of wild-type NRAS was obtained from the Protein Data Bank, and clinically relevant mutations (Gly12, Gly13, Gln61) were introduced computationally. A library natural compounds from the ZINC database was screened, identifying candidates based on binding free energy. Eleven top-ranking compounds were further analyzed for binding interactions, revealing stabilizing hydrogen bonds and hydrophobic contacts, particularly involving residues Phe28 and Asp119. These compounds exhibited strong predicted affinity for the NRAS mutant binding site, representing promising scaffolds for further development. The results demonstrate the utility of computational approaches to accelerate early-stage drug discovery targeting oncogenic NRAS in melanoma.

Keywords: NRAS, melanoma, molecular docking, MEK inhibitor.

1. INTRODUCTION

Melanoma is a deadly form of skin cancer occurring in humans and animals and is responsible for most deaths. At the basal level of the epidermis melanocytes are found, which produce the UV absorbing pigment melanin. (2) It's a combination of environmental and genetic factors, the majority of melanomas arise from somatic mutations acquired later in life. The four major types include Superficial spreading melanoma, Nodular melanoma, Lentigo maligna, Acral lentiginous (2) One of the most known mutated pathways in melanoma that has received a lot of importance for targeted therapeutic purposes, is the MAP kinase signalling pathway. (2) Mutations in each of the identified driver genes lead to deregulation of the MAPK/ERK pathway, leading to uncontrolled cell growth. BRAF subtype show 52% of somatic mutations tumours, following that is NRAS. (4) The average age of diagnosis is 65 but before age 50, the number of women diagnosed are more than men. Among all people with melanoma of the skin, from the time of initial diagnosis, the 5-year survival is 93%. (5) The primary or Stage I melanoma is found only in the skin and is relatively thin. Stage II melanoma extends through the epidermis and further into the dermis. It has a higher chance of spreading through the lymphatic system to a regional lymph node. Stage III is dependent on the size and number of lymph nodes. (6) Stage IV describes melanoma that has spread through the bloodstream to other parts of the body, such as distant locations like soft tissue, distant lymph nodes, or other organs like the lung, liver, brain, bone, or gastrointestinal tract. (6) There have been some clinically approved drugs against melanoma. Trametinib was basically designed for BRAF. But a few

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patients who had mutant NRAS affected cells were tested with this drug. (8) Patients with solid tumours were targeted using Binimetinib. It is a type of MEK inhibitor specifically for mutant NRAS (7). A cysteine residue of post translational modification of RAS are the Farnesyltransferase inhibitors. Initially they were developed for targeting NRAS in combination to other drugs. But since no potential output was obtained, clinical trials have stopped taking place now. (7) Cobimetinib shows a great yield of activity when combined with Vemurafenib. About more than 80% reactivity of BRAF 600-mutant was observed with this therapy. (7) A MEK associated inhibitor. In humans it shows a half way response to NRAS activity while in xenographs it is a therapy for RAS and RAF. Presently under clinical trials. Dabrafenib was developed for treatment against BRAF. (7)
NRAS (Neuroblastoma Ras viral oncogene) gene produces NRAS protein. It acts like a switch for GTP and GDP molecules which is turned On and Off. The N-Ras protein must be turned on by attaching to a molecule of GTP. The N-Ras protein is turned off when it converts the GTP to GDP.When switch is turned off no activation and no signalling is seen by NRAS. This gene belongs to oncogenic family which are cancer causing genes. The proteins produced from NRAS are GTPases. These proteins play important roles in cell division, cell differentiation, cell proliferation. In N-Ras the differences cluster at residues 94–95 (helix 3) and 131–132 (helix 4) (1) A clear difference between isoforms of NRAS is the binding of (RBD) Ras Binding Domain. (1) The G-domain, which catalyses GTP hydrolysis and mediates downstream signalling, is 95% conserved between the Ras isoforms. (1) The Wild Type NRAS has a total of 189 amino acids. The overall structure can be divided into a catalytic G domain, which is comprised (residues 1-86) and (residues 87-166). This is the first crystal structure of NRAS solved at 1.67Å resolution. (1) A clear difference between isoforms of NRAS is the binding of (RBD) Ras Binding Domain. (1) The G-domain, which catalyses GTP hydrolysis and mediates downstream signalling, is 95% conserved between the Ras isoforms. The Wild Type NRAS has a total of 189 amino acids. The overall structure can be divided into a catalytic G domain, which is comprised (residues 1-86) and (residues 87-166). This is the first crystal structure of NRAS solved at 1.67Å resolution. (1) Although targeted inhibitors have been successfully developed for BRAF mutations, effective therapies directly targeting mutant NRAS are lacking, and existing MEK inhibitors have shown only modest clinical success. This unmet medical need motivates the search for novel inhibitors specifically against oncogenic NRAS. In this study, we utilized structure-based virtual screening and molecular docking to explore a comprehensive library of natural compounds from the ZINC database, computationally modeling clinically relevant NRAS mutations at codons 12, 13, and 61. Our screening identified several promising natural compounds with strong predicted binding affinity and stabilizing interactions with critical NRAS residues, providing a valuable starting point for further drug development. The findings underscore the power of computational approaches to accelerate early discovery of specific inhibitors targeting difficult oncogenic drivers in melanoma.

2. EXPERIMENTAL

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2.2. Methods

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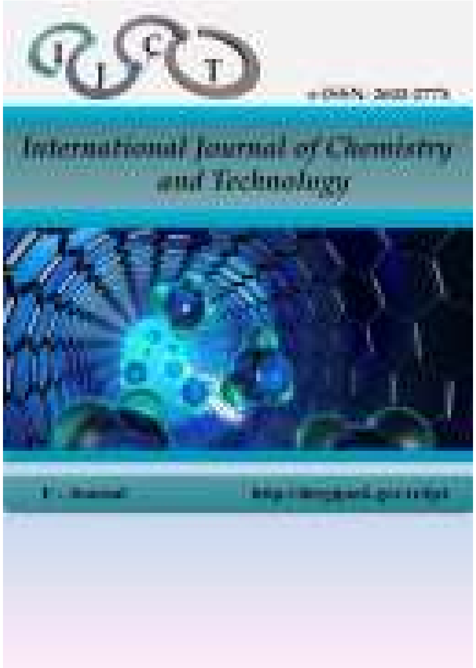

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4.CONCLUSION

Virtual screening enabled the identification of natural compounds with selective inhibitory potential against mutant NRAS variants. Key interactions were observed at both conserved and mutation-specific residues, supporting the structural relevance of the predicted binding modes. Three compounds (ZINC02096813, ZINC02120343, ZINC00518885) displayed the most favorable binding characteristics and consistent stability, indicating their promise as lead molecules for further optimization. These results underscore the value of virtual screening in guiding the discovery of candidate inhibitors for oncogenic NRAS in melanoma..

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Conflict of Interest

The authors declare that they have no conflicts of interest.

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